MSc Final Project · 2026

Predicting Persuasive Comments on r/ChangeMyView

Length, Rhetorical Strategies, and Cross-Corpus Validation

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How Does ChangeMyView Work?



Original View

OP posts an opinion





Reply

Another user challenges it





Delta Awarded

If persuaded,
OP awards a Δ

Can we predict which replies receive a Δ?

Three Research Questions

RQ1 Predictability & Length

Can we predict which comments earn a Delta, and how much does length matter?

RQ2 Rhetorical Strategies

Do rhetorical strategies improve prediction beyond length, and which matter most?

RQ3 Generalization

Do these findings hold on a different dataset, and what changes?

Our Dataset: Predicting Delta Awards

500 discussion posts from r/ChangeMyView

3,971
Comments

641
Delta-awarded comments

16.1%
Positive class

Task: Predict Δ vs. no Δ from comment features

Primary metric: PR-AUC

Initial Approach: A Rich Feature Pipeline

52 Handcrafted features

Reply Features

Linguistic, stylistic, and sentiment signals

Original Post Features

Characteristics of the original viewpoint

Reply–Post Relations

Similarity and contextual relationships

187 Embedding Dimensions

Semantic Embeddings

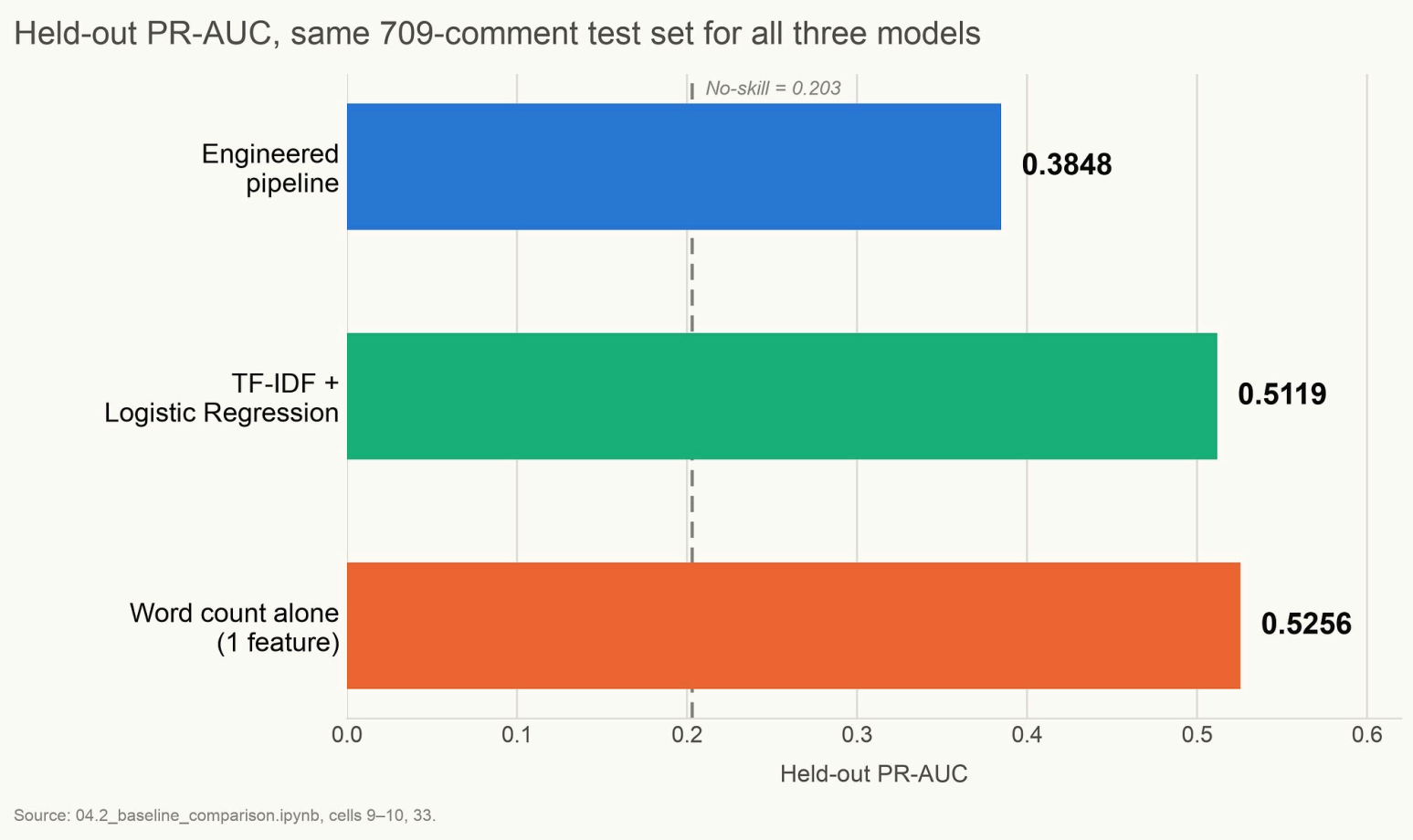
reduced using PCA (85% variance retained)

Combined features matrix

↓

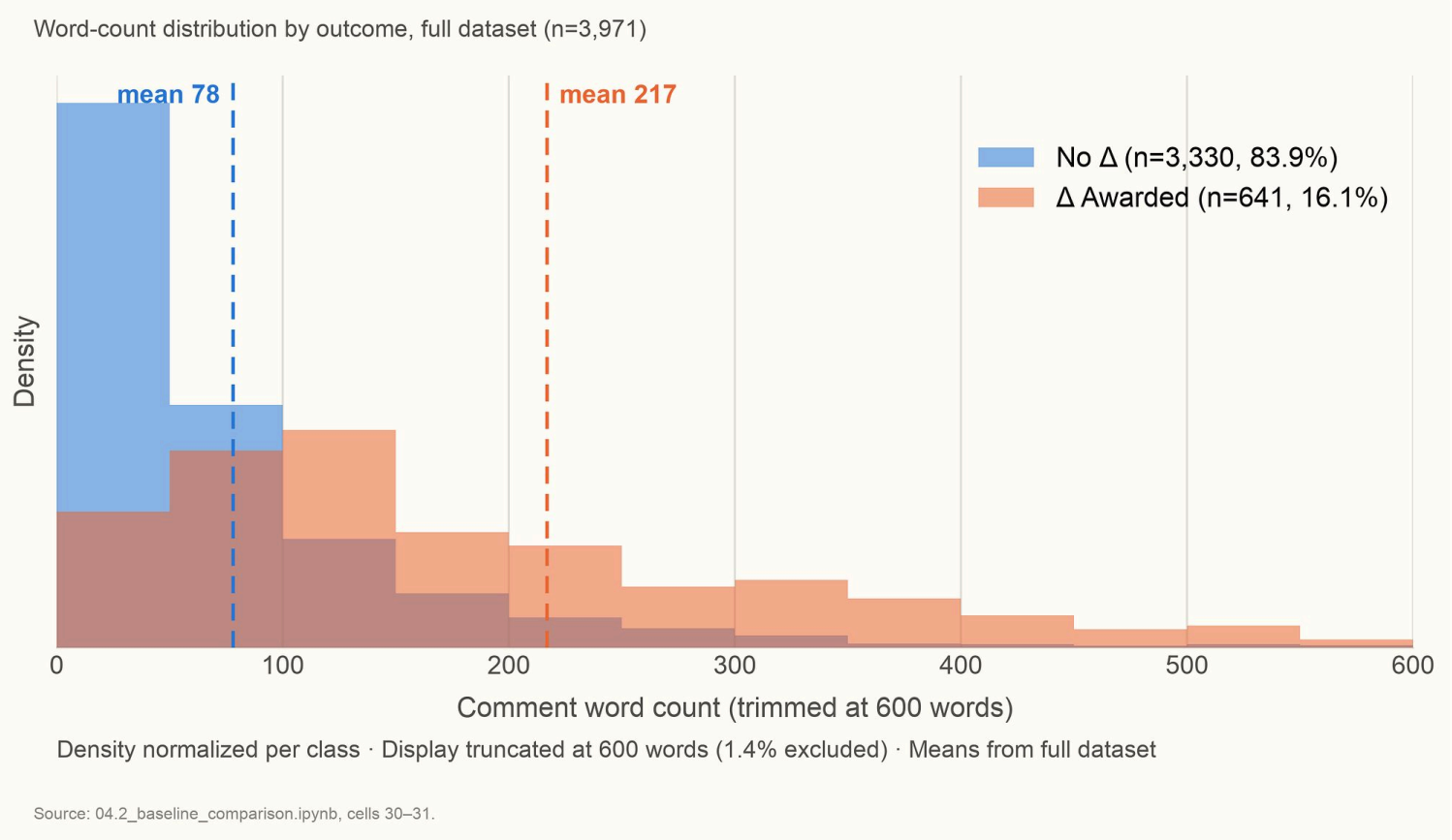
Model Selection → XGBoost

Pivot: Length Beats the Engineered Pipeline



Source: 04.2_baseline_comparison.ipynb, cells 9–10, 33.

Delta-Awarded Comments Are Nearly 3× Longer



No Δ

78

mean words

Δ Awarded

217

mean words

Within-thread: Higher mean word count in 76.5% of mixed-outcome threads.

Eight Rhetorical Strategies

Gemini scored each strategy independently on a 1–5 scale · 2,753 comments



Justification

Giving reasons



Concession

Acknowledging a point



Evidence

Supporting with facts



Refutation

Countering an argument



Personal Narrative

Sharing experience



Simplification

Making ideas clearer



Question Asking

Using questions



Call to Action

Proposing alternatives

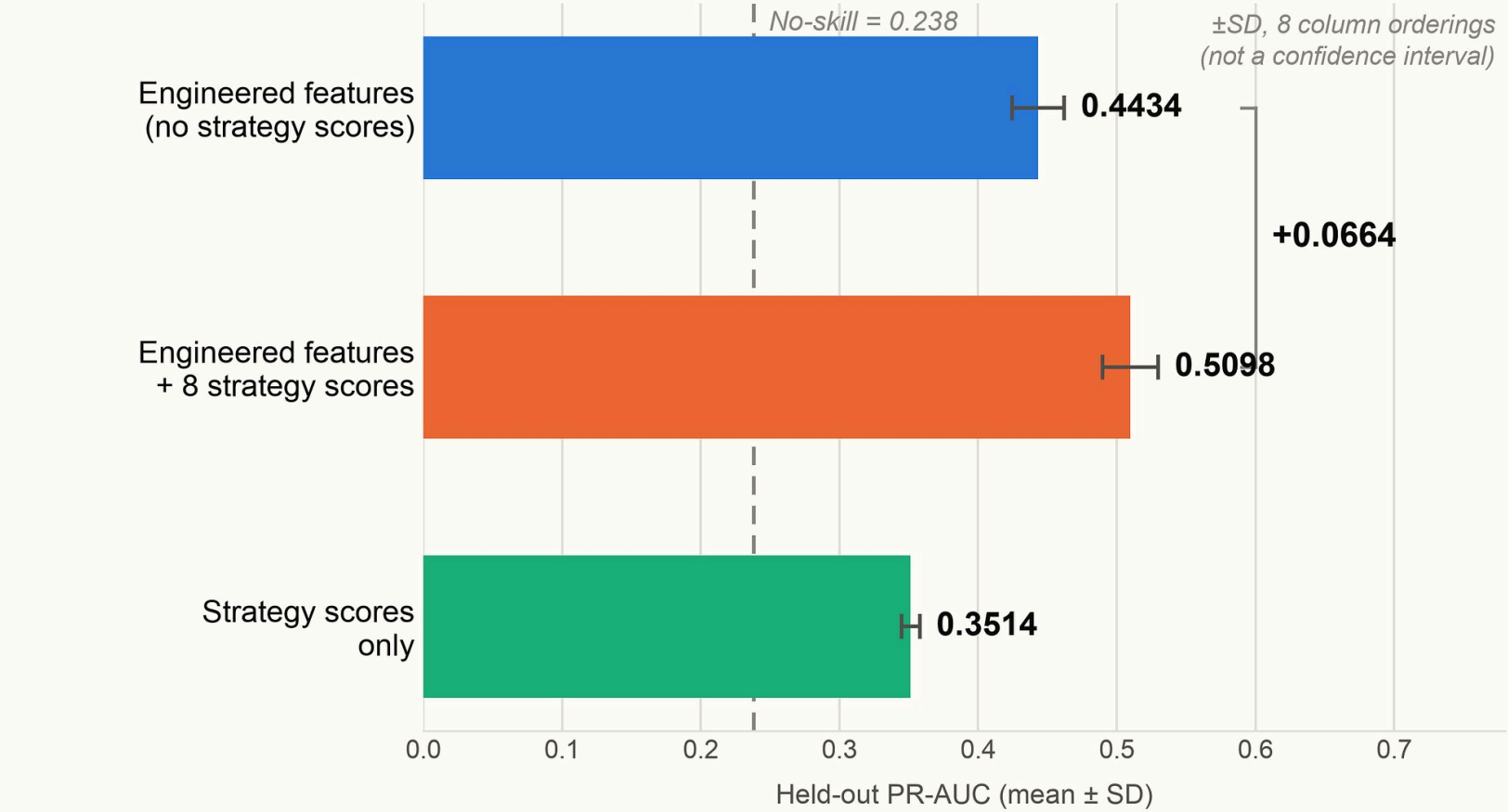
Validation: Test-retest consistency and human rating comparison

Do Rhetorical Strategies Improve Prediction?

+0.0664

PR-AUC improvement

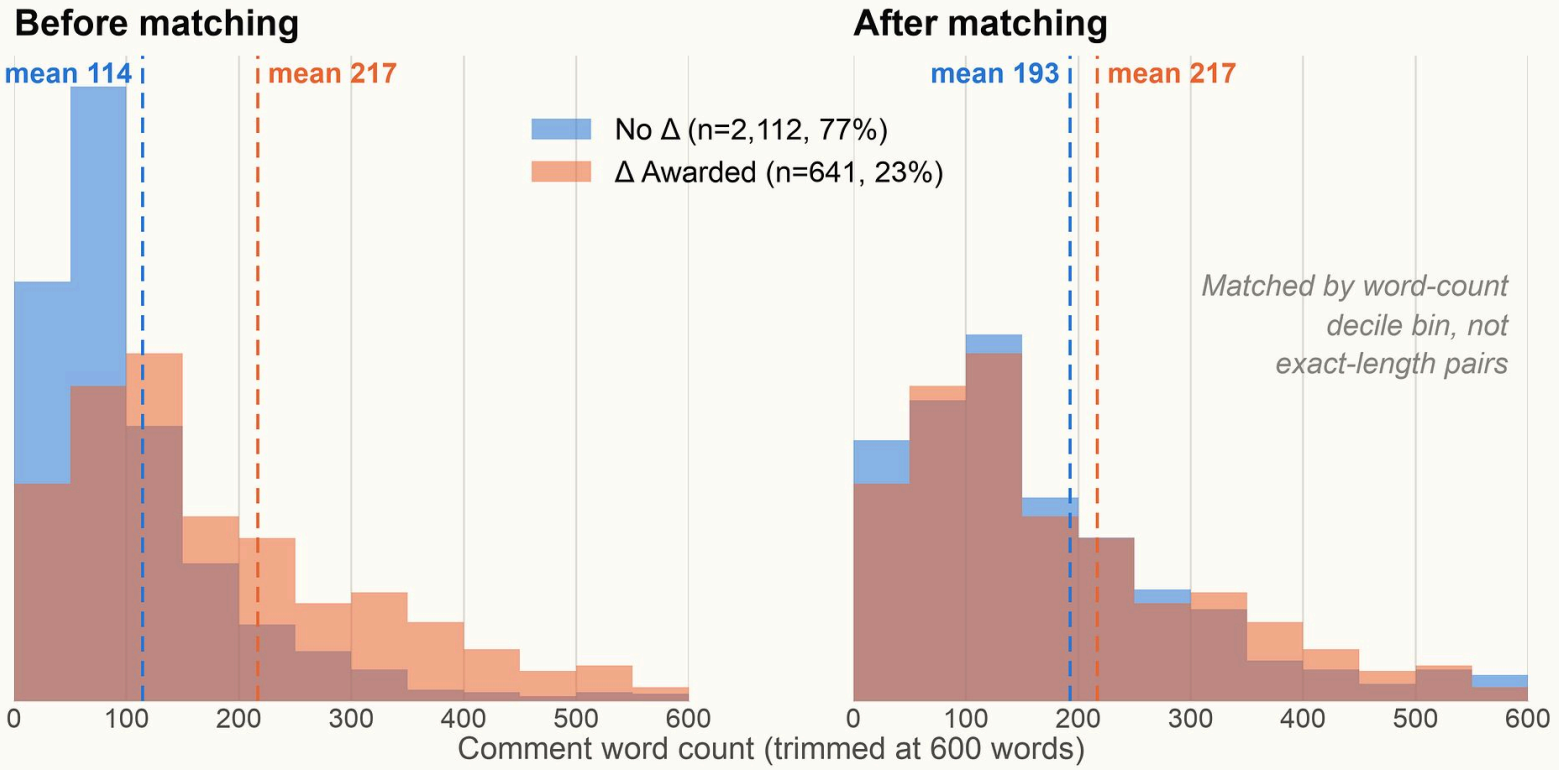
With rhetorical strategy scores



Scored subset, n=2,753 (test: 592 comments)

Source: 06_strategy_ablation.ipynb, cell 17 (headline_df). Column-order sensitivity documented in cells 11–14.

Does the Gain Survive Length Matching?



Descriptive comparison across different samples; not proof of independence from length.

Source: 06_strategy_ablation.ipynb · results/05_cmv_comments_df.csv.zip

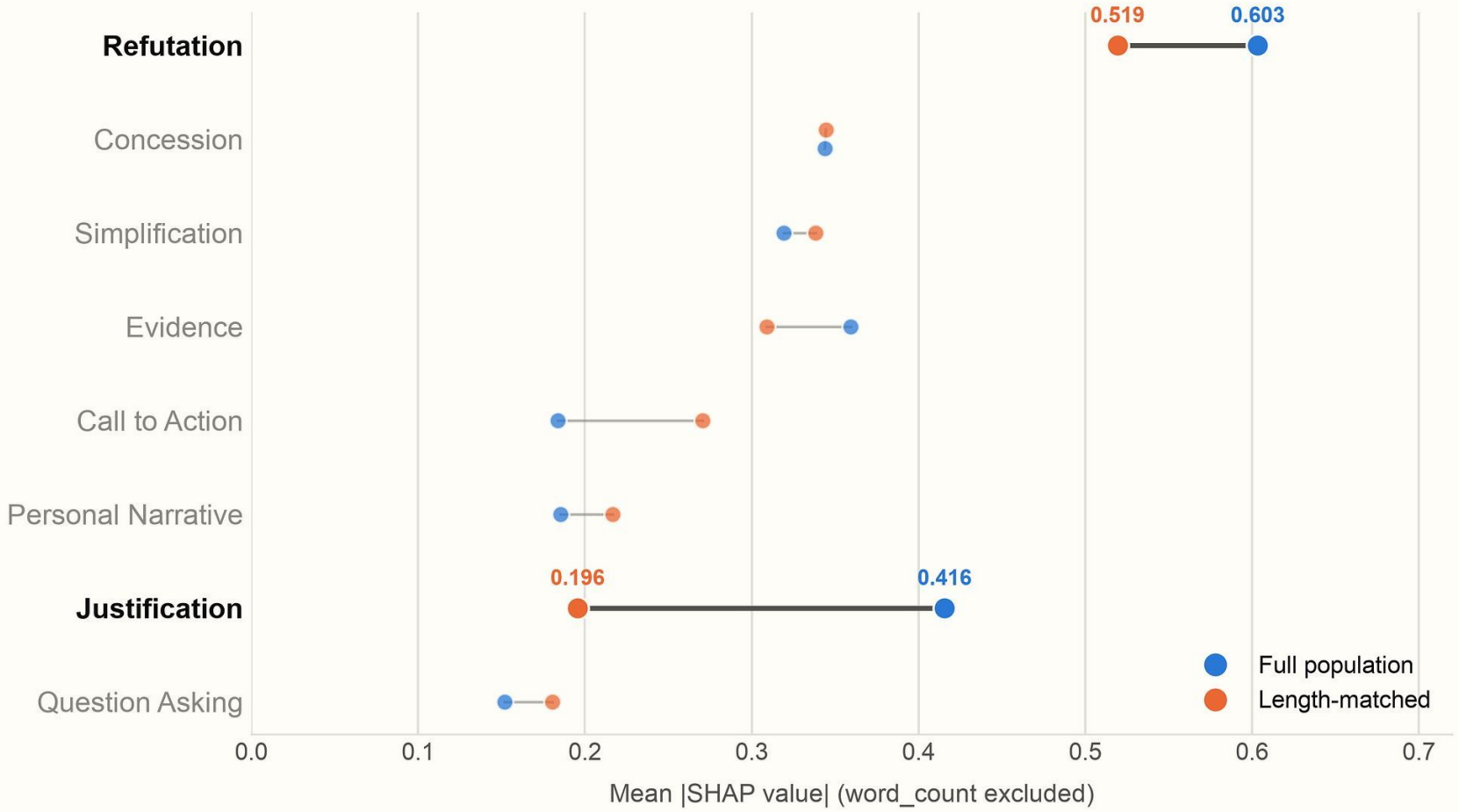
Strategy-score PR-AUC gain

Sample	Δ PR-AUC
Full population (n = 2,753)	+0.0664
Length-matched (n = 1,235)	+0.0557

84%

of predictive gain retained

Refutation Stays On Top After Length Matching



Fitted-model importance, not a causal effect. Source: 07_strategy_interpretation.ipynb, cell 36.

MANN–WHITNEY TEST

3 strategies remain significant after length matching (FDR-corrected):

✔️ Refutation

✔️ Justification

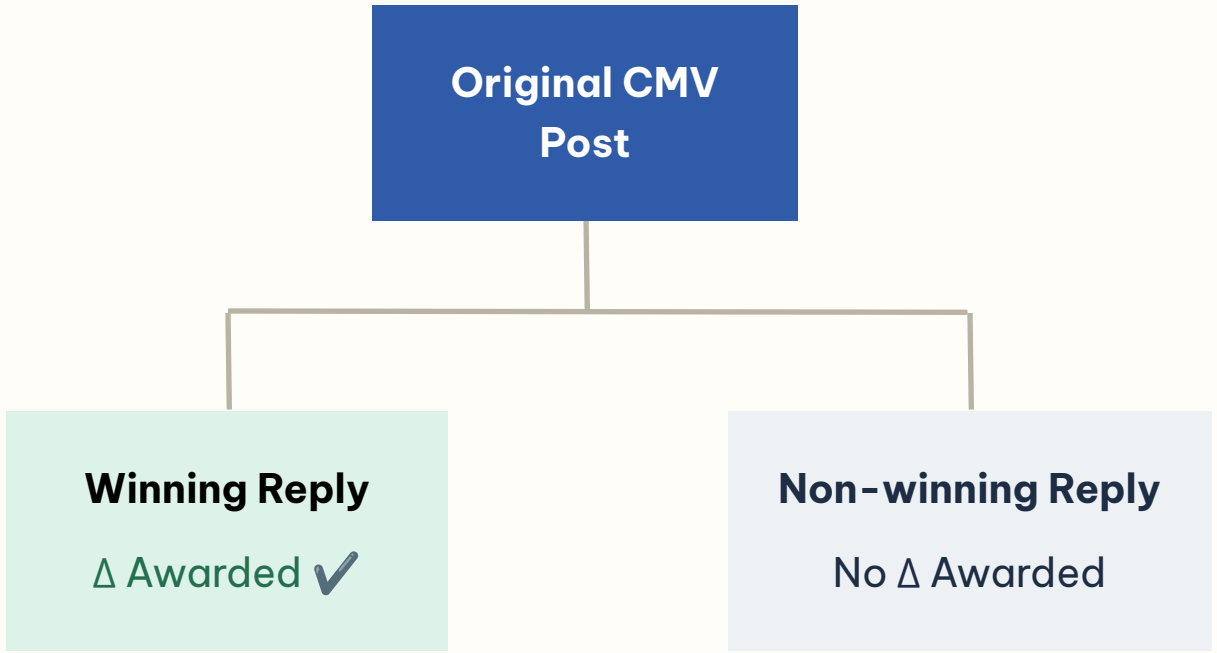
✔️ Simplification

Together with word count, these form the

4-feature Robust-Core model

A Second Dataset: Winning Arguments

Each pair contains two replies to the same discussion.



4,263

Winner-Loser Pairs

Pairwise task: Which reply was more persuasive?

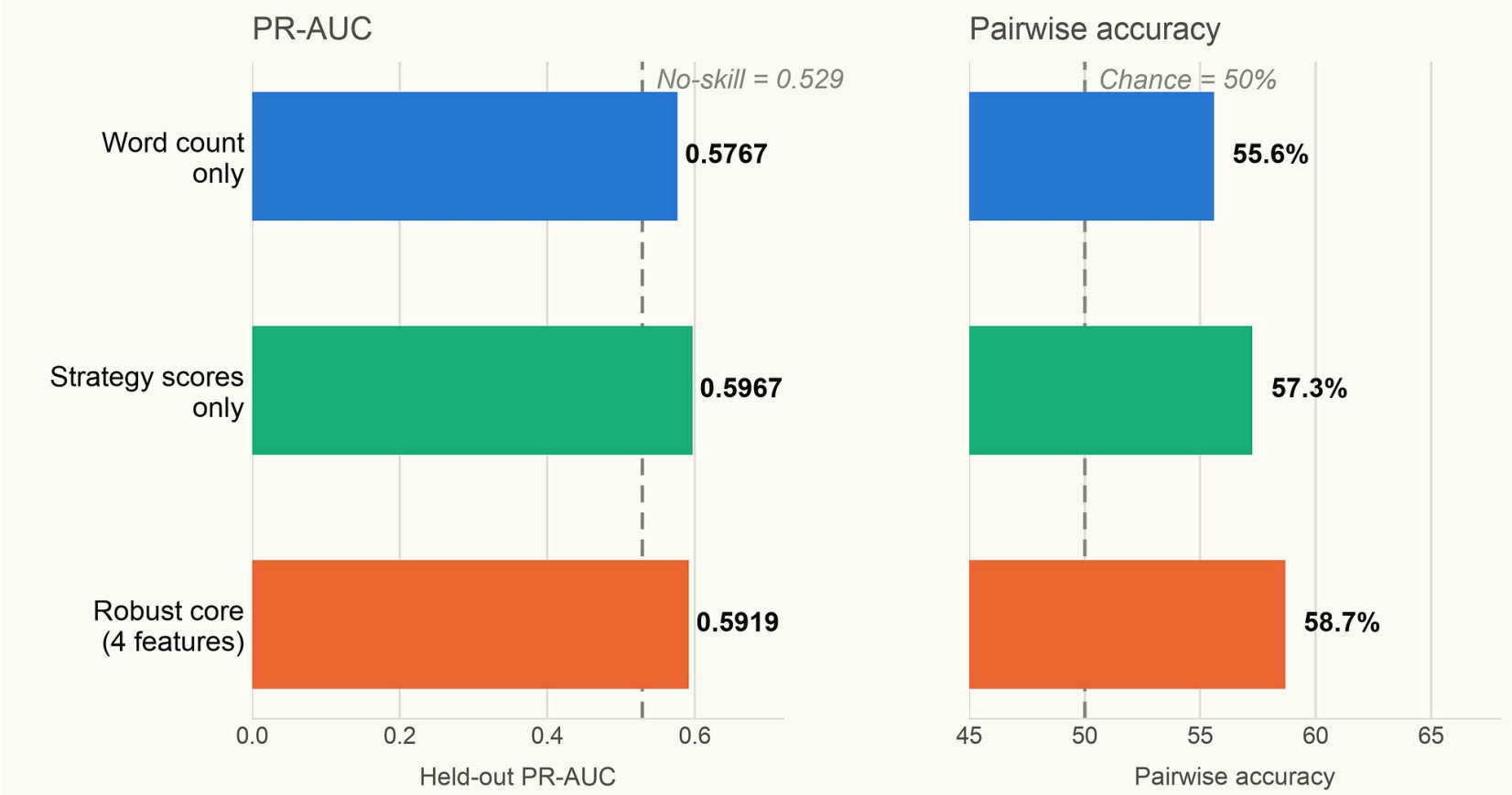
CMV-Selected Strategies Remain Competitive on WA

Retrained on WA

Evaluated on
WA held-out data

Four features, competitive performance

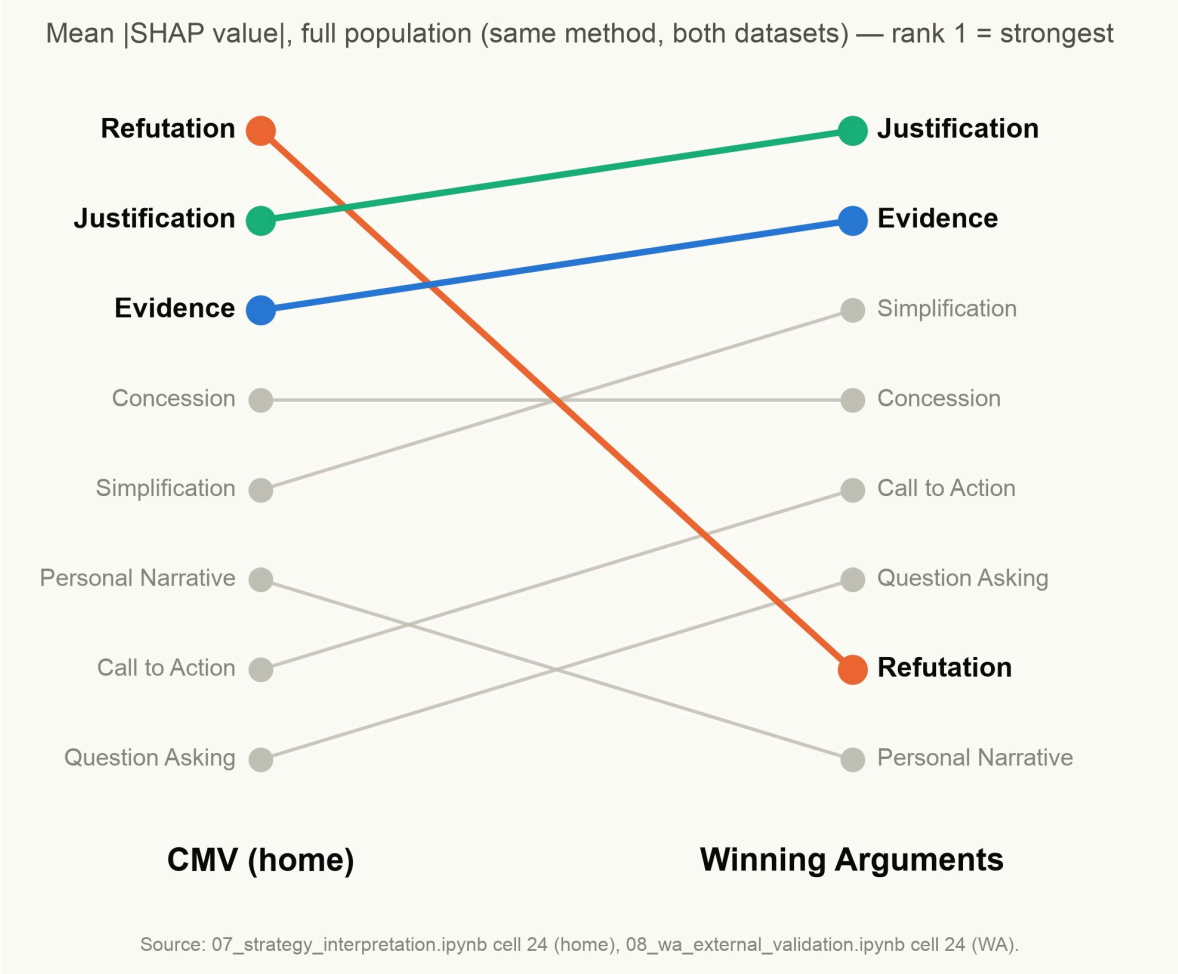
- 3 strategies + word count (vs. 8 strategies)
- PR-AUC close to the full strategy model
- **58.7% pairwise accuracy**, highest of the three



Source: 08_wa_external_validation.ipynb, cells 20, 38–39.

Rhetorical Strategy Importance Shifts Across Datasets

SHAP importance rankings · Full datasets · Independently trained models



Refutation: 1st → 7th

Loses its dominance on WA

Justification: 2nd→ 1st

Becomes the leading strategy

Conclusions: Beyond Length, Context Matters

RQ1

Length Is a Powerful Baseline

Word count alone outperformed a pipeline with over 200 features on our original dataset.

RQ2

Rhetorical Strategies Add Predictive Value

Strategy-score gains largely persisted after length matching.

RQ3

Strategy Rankings Depend on Context

Evidence rose and refutation fell in Winning Arguments.

WHAT I LEARNED?

Strong predictive performance does not necessarily imply meaningful rhetorical insight.

- A simple length baseline challenged the engineered pipeline
- the importance of rhetorical strategies changed across datasets.

What I'd Do Differently?

- Design the length-controlled, cross-dataset evaluation from the start.
- Use a larger human-validated sample

Thank you!

Questions?

Could a 9-Feature Model Replace the Full Pipeline?

Full model · 237 features

PR-AUC: 0.5202 ± 0.0177

Compact model · 9 features

PR-AUC: 0.5379 ± 0.0109

8 strategy scores + word count

Does Model Architecture Matter?

Random Forest

0.5432 ± 0.0065

XGBoost

0.5380 ± 0.0108

Values: mean ± SD across feature-column orderings (not confidence intervals).

Small difference relative to observed variation; **no clear architectural winner.**

Why XGBoost? Kept for consistency with the main experimental pipeline

How Stable Are the PR-AUC Results?

0.0482

PR-AUC variation from changing feature-column order alone

- Same data
- Same model architecture
- Different feature-column order

Why?

Correlated features can have nearly equal split gains, making XGBoost sensitive to feature ordering.

Our response

Report mean ± SD across multiple column orderings for the affected CMV feature comparisons.

Pre-Specified vs. WA-Optimized Features

Pre-specified robust core · 4 features

PR-AUC: 0.5919

Pairwise accuracy: 58.7%

Reported in main results

Best-found variant · 8 features

PR-AUC: 0.5937

Pairwise accuracy: 59.1%

Strategy set without Call to Action

Why report the 4-feature core?

The 4-feature core was specified in advance.

The pre-specified core achieves nearly identical performance without selecting features based on WA results.